

STAT/MA 41600
In-Class Problem Set #22: October 5, 2026
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Problem Set 22 Answers

1a. We have $P(X = Y) = (1/6)P(X = 0) + (1/6)P(X = 1) + \cdots + (1/6)P(X = 5) = (1/6)(P(X = 0) + \cdots + P(X = 5)) = (1/6)(1) = 1/6$.

Alternatively, we can simply recognize that, regardless of the value of X , the random variable Y will agree with X with probability $1/6$. So again we have $P(X = Y) = 1/6$.

1b. We note $P(X > Y) = \sum_{x=1}^5 P(X = x)P(Y < x) = \sum_{x=1}^5 \binom{5}{x} (1/3)^x (2/3)^{5-x} (x/6)$. We compute $(80/243)(1/6) + (80/243)(2/6) + (40/243)(3/6) + (10/243)(4/6) + (1/243)(5/6) = ((80)(1) + (80)(2) + (40)(3) + (10)(4) + (1)(5))/((243)(6)) = 5/18$.

2. Let X_j indicate whether the j th person gets their book back, i.e., $X_j = 1$ if the j th person gets their book back, and $X_j = 0$ otherwise. Of course, we have $X = X_1 + \cdots + X_5$.

2a. We compute $E(X) = E(X_1 + \cdots + X_5) = E(X_1) + \cdots + E(X_5) = 1/5 + \cdots + 1/5 = 1$.

2b. We have $\text{Var}(X) = E(X^2) - (E(X))^2$. We note that $E(X^2) = E((X_1 + \cdots + X_5)^2) = 5E(X_1) + 20E(X_1X_2) = 5(1/5) + 20(1/5)(1/4) = 1 + 1 = 2$. It follows that $\text{Var}(X) = 2 - 1^2 = 1$.

3. Let X_j indicate whether the j th person has 3 bears of the same color, i.e., $X_j = 1$ if the j th person has 3 bears of the same color, and $X_j = 0$ otherwise. Of course, we have $X = X_1 + \cdots + X_{10}$.

3a. We compute $E(X) = E(X_1 + \cdots + X_{10}) = E(X_1) + \cdots + E(X_{10}) = (9/29)(8/28) + \cdots + (9/29)(8/28) = (10)(9/29)(8/28) = 180/203$.

3b. We have $\text{Var}(X) = E(X^2) - (E(X))^2$. We note that $E(X^2) = E((X_1 + \cdots + X_{10})^2) = 10E(X_1) + 90E(X_1X_2) = 10(9/29)(8/28) + 90(9/29)(8/28)((7/27)(6/26)(5/25) + (20/27)(9/26)(8/25)) = 4320/2639$. It follows that $\text{Var}(X) = 4320/2639 - (180/203)^2 = 455760/535717 = 0.8507$.

4. The probability that the 2 red beads are placed together is $2/8$ because regardless of where the first red bead goes, there are 2 out of the remaining 8 places for the other red bead. The probability that all of the blue beads will be immediately next to the reds clockwise is $(3/7)(2/6)(1/5)$, and similarly, the probability that all of the blue beads will be immediately next to the reds counterclockwise is $(3/7)(2/6)(1/5)$. If the red beads are together and the blue beads are together, then the yellow beads will be together too. So the desired probability is $(2/8)(2)(3/7)(2/6)(1/5) = 1/70$.

5. Each piece of candy survives until the point in question (i.e., until the bag only contains candies of one color) if and only if it is eaten after all of the candies of the other color, which happens with probability $1/11$. So the expected number of candies that remain at the end is $(20)(1/11) = 20/11 = 1.8182$.